Nisha Patel

PhD Researcher — Urban Climate Modelling & Remote Sensing, Climate Data Science

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Languages: English (fluent), German (B2), Gujarati (native), Hindi (fluent)
Nationality: German

PROFILE

Atmospheric scientist and PhD researcher with 8+ years of experience across meteorology, remote sensing, and urban climate modelling. Skilled in translating large environmental datasets and machine-learning methods into actionable insights for climate-resilient city planning. Experienced in both academic research and applied data analysis, with a record of peer-reviewed publications, international collaboration, and teaching.

ACADEMIC RECORDS

PhD, Urban Climate Modelling: *University of Kassel, Germany*

Jun 2022 – Present

Research focus: climate adaptation in cities, climate-resilient urban design, urban climate modelling, and machine-learning methods for upscaling microclimate models.

M.Sc., Meteorology: *Karlsruhe Institute of Technology & Leibniz Universität Hannover, Germany*

Oct 2019 – May 2022

Coursework: Data Analysis & Machine Learning, Turbulent Diffusion, Remote Sensing, ICON Modelling, Cloud Dynamics, Tropical Meteorology, Radiation, IPCC Seminar, Atmosphere–Ocean Interaction, Theoretical Meteorology.

Thesis: Determined Global Horizontal Irradiance from satellite data (Meteosat-2) and validated results against ground-based measurements.

M.Sc., Physics: *M.M. Patel Institute of Science and Research, Gandhinagar, Gujarat, India*

Apr 2015 – Jun 2017

Coursework: Classical Mechanics, Theoretical Physics, Space & Astronomy, Atmospheric Science, Nuclear Physics.

Thesis: Investigated atmospheric dynamics using temperature and ozone parameters from SABER satellite data (2008–2018).

B.Sc., Physics: *Gujarat University, Ahmedabad, Gujarat, India*

Apr 2012 – Mar 2015

Coursework: Classical Mechanics, Theoretical Physics, Atmospheric Science, Nuclear Physics, Optics.

CERTIFICATIONS

- **Google Data Analytics Professional Certificate**: Google Career Certificates (Oct 2021 – Apr 2022)
- **Climate-Related Financial Risk Act (SB 261)**: Imperial Business School (May, 2026)

WORK EXPERIENCE

Research Associate (Wissenschaftliche Mitarbeiterin): *University of Kassel, Germany*

Jun 2022 – Present

- Design and run micro-climate simulations (ENVI-met) to evaluate heat-mitigation and climate-adaptation strategies for urban neighbourhoods.
- Develop machine-learning models to upscale point-based micro-climate simulations to city-wide predictions.
- Collect and analyse field-measurement data on urban micro-climates across Kassel.
- Co-authored a peer-reviewed review paper on indicators for heat- and water-sensitive urban development, published 2025.

Tools: *Python, ML frameworks, ENVI-met, Tableau*

Student Assistant (HiWi): *Karlsruhe Institute of Technology (KIT), Germany*

Dec 2019 – Dec 2020

- Managed and processed ice-nucleating particle (INP) sample databases, maintaining data integrity for ongoing research.
- Extracted and analysed INP data, producing daily, monthly, and yearly time-series plots of temperature and INP concentration.
- Presented findings regularly to the research group, supporting the team's atmospheric aerosol research.

Tools: *Python, MATLAB, LabVIEW, SQL*

Project Associate*: Indian Institute of Technology Madras, India*

Feb 2019 – Aug 2019

- Ran Weather Research and Forecasting (WRF) model simulations using ECMWF and NCEP reanalysis datasets.
- Applied the GEOS-Chem chemical-transport model to support atmospheric-composition analysis.
- Carried out data analysis supporting air-quality and atmospheric-chemistry research.

Tools: *Python, MATLAB, GrADS, NCL, Clima-view, Linux*

Student Visitor*: Physical Research Laboratory (PRL), Dept. of Space & Atmospheric Science, India*

Apr 2017 – Feb 2019

- Collected and visualised ground-based atmospheric data from Raman Lidar, Rayleigh Lidar, and Ceilometer instruments.
- Supported instrument operation and data-quality control for space and atmospheric-science research.

Tools: *LabVIEW, MATLAB, MS Excel*

TEACHING EXPERIENCE

Teaching Assistant *: University of Kassel, Germany*

Jun 2022 – Present

- Deliver micro-climate simulation training using ENVI-met (Winter Semester).
- Lead micro-climate measurement and data-analysis practicals for Kassel field campaigns (Summer Semester).

Science Educator *: Vikram A. Sarabhai Community Science Centre (VASCSC), India*

Jun 2017 – Jun 2018

- Taught physics, astronomy, and rocketry to students across the School Science Forum and summer programmes.
- Mentored student project groups through hands-on laboratory learning.

SKILLS & SOFTWARE

Programming & Data Analysis: Python, MATLAB, SQL, R

Modelling & GIS: ENVI-met, WRF, QGIS, Arc-GIS

Visualization & Reporting: Tableau, MS Office

PUBLICATIONS

Patel, N., Mackey, L., Zhong, J., Lu, Y., Jaenicke, B., Burghardt, R., Pericleous, K. *Towards Rapid and Data-Efficient Urban Climate Assessment using Machine Learning Frameworks*. **Computational Urban Science**. Springer. [Submitted]

Otto, F., Burghardt, R., Jänicke, B., **Patel, N.** (2026). Entwicklung von messbaren Zielen, Indikatoren und Maßnahmen der Klimaanpassung durch Stadtentwicklung. In: Altrock, U., et al. Stadterneuerung und Klimawandel. Jahrbuch Stadterneuerung. Springer VS, Wiesbaden. https://doi.org/10.1007/978-3-658-49473-5_14

Patel, N., Jänicke, B., Burghardt, R., Vulova, S., Otto, F. (2025). Assessing progress in urban climate adaptation: A review of indicators for heat and water-sensitive urban development. Climate Resilience and Sustainability. https://doi.org/10.1002/cli2.70009

Patel, N., Sharma, S., Joshi, V., Kumar, P., Ojha, N., Kumar, K. N., Chandra, H., Beig, G. (2019). Observations of Middle Atmospheric Seasonal Variations and Study of Atmospheric Oscillations at Equatorial Regions. Journal of Atmospheric and Solar-Terrestrial Physics. https://doi.org/10.1016/j.jastp.2019.105066

Joshi, V., Sharma, S. K., Kumar, K. N., **Patel, N.,** Kumar, P., Ghosh, P., Bencherif, H., Jethva, C., Vaishnav, R. (2020). Analysis of the Middle Atmospheric Ozone using SABER Observations: A study over Mid-latitudes in the Northern and Southern Hemisphere. Climate Dynamics. DOI: 10.1007/s00382-020-05124-6

SCHOLARSHIPS & GRANTS

- Multi-scale, Multi-disciplinary, Modelling for impact' programme (M34impact) Research Scholarship (2026)
- Indo-German Centre for Sustainability, DAAD Award (2024)
- ASL International Scholarship for Doctoral Students (2023)
- World Landscape Architecture Honour Award (2023)